

WEEK 5 PROJECT SUMMARY

During Week 5, the HealthConnect Data Analytics track moved from the planning and foundation stage established in Week 4 into practical data preparation, exploratory analysis, KPI development, and business insight generation. The work continued to address the main HealthConnect project question: **How can HealthConnect Clinic use data and AI to reduce missed appointments and improve the patient support experience?**

The HealthConnect Appointment Dataset was used as the primary source for the analysis. The dataset contains 5,000 appointment records and 18 variables, covering appointments from 1 January 2025 to 30 June 2026. The dataset includes information about patient demographics, appointment characteristics, booking behaviour, previous appointment history, previous no-shows, reminders, distance to the clinic, waiting time, and appointment outcomes.

During data preparation, no duplicate appointment records were identified, and the appointment IDs were unique. Some missing values were identified, including 90 missing distance-to-clinic values, 60 missing waiting-time values, and 1,366 missing reminder-channel values. The missing reminder-channel values were generally consistent with appointments where no reminder was sent. These missing values were documented rather than automatically replaced to avoid introducing unsupported assumptions into the analysis.

A business-rule inconsistency was also identified. The dataset contains appointments scheduled on Sundays, even though the HealthConnect Clinic Knowledge Base states that the clinic is closed on Sundays. This issue should be clarified before Sunday-related findings are used to support operational decisions.

The Week 5 analysis produced five important KPIs: Total Appointments, Attendance Rate, No-Show Rate, Cancellation Rate, and Reminder Coverage Rate. The dataset contained 5,000 appointments, with 2,423 No-Shows (48.46%), 2,314 Attended appointments (46.28%), and 263 Cancelled appointments (5.26%). The overall reminder coverage rate was 72.68%.

The exploratory analysis identified several important patterns associated with appointment attendance and no-shows. Previous no-show behaviour showed the strongest pattern. The no-show rate increased from 43.51% for patients with no previous no-shows to 68.82% for patients with three or more previous no-shows. This suggests that previous attendance behaviour may be an important indicator of future appointment attendance.

Booking lead time also showed a strong relationship with no-shows. Appointments booked 31 or more days in advance had a 60.49% no-show rate, compared with 26.51% for appointments booked 0–1 days in advance. This suggests that appointments scheduled far in advance may require additional communication or engagement strategies.

Reminder status showed another important difference. Appointments where a reminder was sent had a 47.36% no-show rate, compared with 51.39% where no reminder was sent. This indicates a potential relationship between reminder coverage and attendance. However, the result represents an association and does not prove that reminders directly caused patients to attend their appointments.

Appointment type also showed differences in no-show rates. Follow-up appointments had the highest no-show rate at 51.23%, followed by Diagnostic Tests at 49.75%, Specialist Consultations at 47.44%, and General Consultations at 46.64%. This suggests that follow-up appointments may require additional investigation and targeted patient-support strategies.

Distance to the clinic was another potential factor. Patients living more than 20 km from the clinic had a 57.76% no-show rate, suggesting that accessibility and travel-related challenges may contribute to missed appointments. Further analysis would be required before drawing operational conclusions.

Based on these findings, several recommendations were developed for HealthConnect. The clinic should prioritise additional support for patients with previous no-shows, particularly patients with three or more previous no-shows. HealthConnect should also strengthen communication and reminders for appointments booked 31 or more days in advance and work toward improving the current 72.68% reminder coverage.

The effectiveness of reminders should continue to be evaluated before assuming that reminders directly cause improved attendance. HealthConnect should also investigate possible accessibility challenges for patients who live far from the clinic and examine why follow-up appointments have a relatively high no-show rate. In addition, the Sunday operating-day inconsistency between the dataset and Clinic Knowledge Base should be resolved.

The Week 5 findings were implemented visually through a Microsoft Power BI dashboard. The dashboard provides an overview of appointment performance using five KPI cards: Total Appointments, Attendance Rate, No-Show Rate, Cancellation Rate, and Reminder Coverage. It also includes visualisations for no-show rates by previous no-shows, booking lead time, reminder status, appointment type, and distance to the clinic, as well as the overall appointment outcomes.

The analysis has several limitations. Missing values exist in distance-to-clinic, waiting-time, and reminder-channel fields. The Sunday operating-day inconsistency also requires clarification. Most importantly, the analysis identifies associations rather than causal relationships. For example, the lower no-show rate among patients who received reminders does not prove that reminders caused patients to attend.

The dataset is also fictional and anonymised. Therefore, the findings should not automatically be generalised to real healthcare organisations without further validation using appropriate real-world data.

The Week 5 analysis also provides useful inputs for future collaboration with the Data Science track. Variables such as **previous no-shows, booking lead time, reminder status, reminder channel, appointment type, distance to the clinic, and other appointment characteristics** could potentially be considered for future predictive modelling. These variables should first be assessed for data quality, predictive value, and possible data leakage.

Overall, Week 5 successfully moved the HealthConnect Data Analytics project from planning into practical implementation. Analysis of the **5,000-record HealthConnect Appointment Dataset** produced important KPIs, a Power BI dashboard, and several business insights. The findings particularly highlight **previous no-show history and booking lead time** as important areas for further investigation, while reminder coverage, appointment type, and distance to the clinic provide additional opportunities for analysis.

For Week 6, the project should focus on conducting deeper analysis, validating the strongest patterns, refining the Power BI dashboard, investigating high-risk appointment groups, examining reminder effectiveness in greater detail, and resolving outstanding data-quality and business-rule issues. These activities will help HealthConnect develop more targeted, evidence-based strategies for improving appointment attendance and patient support while providing stronger analytical inputs for future predictive analysis.